

CSE461: Introduction to Robotics

Hardware Project Proposal — Three Project Ideas

Group: _____ Members & IDs: _____ Date: _____

Guideline Compliance Summary

All three ideas are fully autonomous, problem-solving systems (no line-following or remote-controlled robot is proposed as a standalone project). Each idea meets the minimum of one microcontroller, a chassis/frame, a dedicated power supply, at least 3 sensor types, and at least 2 actuator types.

Idea	Autonomous loop	Sensor types	Actuator types	Cost (Tk.)
1. SmartSort	Sense → classify (on-device CNN) → actuate; no human handling	3 (camera, IR proximity, ultrasonic)	3 (servos, exhaust fan, buzzer/LED)	5,469
2. AirSense	Autonomous grid coverage, on-board classification and response	5 (gas, dust, temp/RH, ultrasonic, wheel encoders)	3 (drive motors, sampling fan, buzzer/LED)	5,184
3. AgniGuard	Patrol → verify → suppress → alert, fully unattended	4 (flame x3, smoke, temperature, ultrasonic)	4 (drive motors, pump, servo, buzzer)	3,279

Project Idea 1

1. Project Title

SmartSort — AI-Based Automatic Waste Segregation Robot

2. Purpose

Objective: To design a compact, bin-side robotic sorting system that autonomously identifies each dropped solid-waste item as biodegradable or non-biodegradable using an on-device machine learning model, and physically routes it into the correct compartment — with a confidence-gated reject path for ambiguous items — without any human handling at any stage.

Scope: Households, dormitory common rooms, university corridors, and small offices in Dhaka, where source-level segregation is almost never practiced. The system is a stationary robotic platform implementing a complete autonomous perception–decision–actuation loop (sense: camera/IR/ultrasonic; decide: on-device CNN; act: servos/fan/buzzer), rather than a mobile robot.

Significance: Dhaka generates roughly 6,500 tonnes of municipal solid waste per day, of which approximately 70–80% is organic; because segregation at the source is rare, compostable material is landfilled at Matuail and Aminbazar instead of being recovered. A replicable sorting bin costing under Tk. 6,000 demonstrates embedded machine learning solving this problem at the exact point of disposal, and additionally yields usage data (classification counts, reject rate, fill rates) useful for studying household waste behaviour.

3. Components

- **Microcontrollers:** ESP32-CAM (AI-Thinker, 4 MB PSRAM) for image capture and on-device TensorFlow Lite Micro inference, paired with an Arduino Uno for sensor/actuator control. The two boards communicate over UART, with a resistor divider on the Uno-TX → ESP32-RX line (5 V → 3.3 V logic).

- **Sensors (3 types):** ESP32-CAM onboard OV2640 camera (waste classification); IR proximity sensor (item-presence trigger at the imaging stage); 2× HC-SR04 ultrasonic sensors (per-compartment fill-level monitoring).
- **Actuators (3 types):** 2× MG90S metal-gear micro servos — one entry-gate servo that doubles as the imaging stage, and one three-position sorting-flap servo (left = biodegradable, right = non-biodegradable, centre = reject tray); one 5 V exhaust fan for odor control on the biodegradable compartment; buzzer + LED for full-bin and reject alerts.
- **Body/Chassis:** Single-chute plywood/acrylic housing with two main compartments plus a narrow centre reject tray; matte-white, LED-illuminated imaging chamber; 3D-printed or laser-cut gate and flap brackets.
- **Additional Components:** 18650 Li-ion pack with charging module feeding a 5 V/2 A buck converter with $\geq 470\ \mu\text{F}$ bulk capacitance (prevents the well-known ESP32-CAM brownout during flash capture); common ground across boards; breadboard and jumper wires.

4. Cost Breakdown (Approx.)

Components	Unit Cost (Tk.)	Quantity	Total (Tk.)
ESP32-CAM (AI-Thinker) module	940	1	940
Arduino Uno R3	939	1	939
MG90S Metal-Gear Micro Servo	180	2	360
Ultrasonic Sensor (HC-SR04)	125	2	250
IR Proximity Sensor	70	1	70
5 V Exhaust Fan (odor control)	150	1	150
White LED Ring/Strip (imaging chamber)	60	1	60
Buzzer + LED Module	30	1	30
18650 Battery Pack + Charging Module	300	1	300
5 V / 2 A Buck Converter (LM2596/MP1584)	120	1	120
Acrylic/Plywood Chassis & Bin Compartments	1200	1	1200
Jumper Wires + Breadboard	350	1	350
Miscellaneous (screws, glue, wiring, capacitors, divider resistors)	700	1	700
Estimated Total			5,469 Tk.

5. Functionality Breakdown

- **Functionality 1: Controlled Image Capture and On-Device Classification**
 - **Overview:** Each item is held on the closed entry gate inside an LED-lit imaging chamber, photographed at a fixed distance against a fixed background, and classified directly on the ESP32-CAM.
 - **Working Procedure:** The IR sensor detects an item resting on the gate and triggers the ESP32-CAM to capture a frame under constant LED illumination. An int8-quantised CNN (MobileNet-class, $\sim 96 \times 96$ input) trained via Edge Impulse on a closed set of 10–12 common household waste items (150–300 images per item, captured with this same camera inside this same chamber) outputs a biodegradable/non-biodegradable label with a softmax confidence, sent to the Uno over UART. The controlled imaging domain — fixed optics, lighting, and

background — is what makes $\geq 85\%$ held-out accuracy realistic on a microcontroller; inference takes roughly 1–2 s per frame.

- **Functionality 2: Confidence-Gated Three-Way Physical Sorting**

- **Overview:** Mechanically routes each item by predicted class, and deliberately refuses to guess on ambiguous items.
- **Working Procedure:** If confidence ≥ 0.80 , the Uno pre-positions the sorting flap to the corresponding side and opens the gate servo; the item slides into the correct compartment and both servos reset for the next cycle (≤ 5 s per item, items up to ~ 200 g). If confidence < 0.80 , the flap centres and the item drops into a small manual reject tray with a short buzzer chirp. Misclassification risk is thereby converted into a designed, observable behaviour instead of contamination of the wrong bin.

- **Functionality 3: Fill-Level Monitoring, Odor Control, and Wi-Fi Usage Dashboard**

- **Overview:** Supervises both compartments, manages odor, and reports live usage analytics at zero additional component cost.
- **Working Procedure:** An HC-SR04 above each compartment samples every 30 s using a median-of-5 filter (suppresses multipath echoes inside the narrow bin); when headroom falls below a preset threshold, the buzzer and LED signal a full bin. The exhaust fan runs on a duty cycle whenever the biodegradable compartment is non-empty. The ESP32-CAM's built-in Wi-Fi pushes classification counts, reject rate, and fill levels to a ThingSpeak/Firebase dashboard, making the bin 'smart-city ready' with no extra hardware.

6. Feasibility Notes and Performance Targets

- **Declared operating envelope:** one item per cycle, ≤ 200 g, ≥ 3 cm minimum dimension, drawn from the trained closed item set.
- **Performance targets:** $\geq 85\%$ binary accuracy on a held-out image set; $\geq 95\%$ routing accuracy for predictions with confidence ≥ 0.80 ; ≤ 5 s cycle time; fill-level alert accurate to ± 2 cm.
- **Key risks and mitigations:** ESP32-CAM brownout during capture $\rightarrow$ 5 V/2 A buck plus bulk capacitance; servo stall under item impact $\rightarrow$ metal-gear MG90S with the flap hinged near its centroid; open-world inputs $\rightarrow$ closed item set plus the confidence-gated reject path; UART logic-level mismatch $\rightarrow$ resistor divider on the Uno-TX line.